Arithmetic Expression Evaluator in C++

Test Cases

Version 1.0

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Revision History

Date	Version	Description	Author
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Test Cases

1. Purpose

This Test Case Specification document for the Calculatorz defines a test case for an item that should be tested.

<The sections of the test case specification shall be ordered in the specified sequence. Additional sections may be included at the end. If some or all of the content of a section is in another document, then a reference to that material may be listed in place of the corresponding content. The referenced material must be attached to the test case specification or available to users of the case specification.>.

2. Basic Valid Expression

2.1 Test case identifier

ID: TC-TOK-01

2.2 Test item

Tokenizer module. This test checks whether the tokenizer correctly identifies parentheses, numbers, addition, and division operators in a normal arithmetic expression.

2.3 Input specifications

Input: (3 + 5) / 6

2.4 Output specifications

Output: <{LeftParen "("}, {Number "3"}, {Plus "+"}, {Number "5"}, {RightParen ")"}, {Divide "/"}, {Number "6"}>

2.5 Environmental needs

Specify any other requirement.

2.6 Special procedural requirements

Describe any special constraints on the test procedures that execute this test case. These constraints may involve special set up, operator intervention, output determination procedures, and special wrap up.

2.7 Intercase dependencies

List the identifiers of test cases that must be executed prior to this test case. Summarize the nature of the dependencies.

3. Whitespace Handling

3.1 Test case identifier

ID: TC-TOK-02

3.2 Test item

Tokenizer module. This test checks whether the tokenizer ignores extra spaces and still produces the correct tokens.

3.3 Input specifications

Input: 10 * 2

3.4 Output specifications

Output: <{Number "10"}, {Multiply "*"}, {Number "2"}>

3.5 Environmental needs

Specify any other requirement.

3.6 Special procedural requirements

Describe any special constraints on the test procedures that execute this test case. These constraints may involve special set up, operator intervention, output determination procedures, and special wrap up.

3.7 Intercase dependencies

List the identifiers of test cases that must be executed prior to this test case. Summarize the nature of the dependencies.

4. Multi-digit Numbers

4.1 Test case identifier

ID: TC-TOK-03

4.2 Test item

Tokenizer module. This test checks whether the tokenizer groups multiple digits into one number token instead of separating each digit.

4.3 Input specifications

Input: 123 + 456

4.4 Output specifications

Output: <{Number "123"}, {Plus "+"}, {Number "456"}>

4.5 Environmental needs

Specify any other requirement.

4.6 Special procedural requirements

Describe any special constraints on the test procedures that execute this test case. These constraints may involve special set up, operator intervention, output determination procedures, and special wrap up.

4.7 Intercase dependencies

List the identifiers of test cases that must be executed prior to this test case. Summarize the nature of the dependencies.

5. Decimal Numbers

5.1 Test case identifier

ID: TC-TOK-04

5.2 Test item

Tokenizer module. This test checks whether the tokenizer correctly recognizes floating-point numbers.

5.3 Input specifications

Input: 3.14 * 2.5

5.4 Output specifications

Output: <{Number "3.14"}, {Multiply "*"}, {Number "2.5"}>

5.5 Environmental needs

Specify any other requirement.

5.6 Special procedural requirements

Describe any special constraints on the test procedures that execute this test case. These constraints may involve special set up, operator intervention, output determination procedures, and special wrap up.

5.7 Intercase dependencies

List the identifiers of test cases that must be executed prior to this test case. Summarize the nature of the dependencies.

6. Invalid Character

6.1 Test case identifier

ID: TC-TOK-05

6.2 Test item

Tokenizer module. This test checks whether the tokenizer detects unsupported characters that are not numbers, operators, or parentheses.

6.3 Input specifications

Input: 5 + a

6.4 Output specifications

Output: Error: Invalid token "a"

6.5 Environmental needs

Specify any other requirement.

6.6 Special procedural requirements

Describe any special constraints on the test procedures that execute this test case. These constraints may involve special set up, operator intervention, output determination procedures, and special wrap up.

6.7 Intercase dependencies

List the identifiers of test cases that must be executed prior to this test case. Summarize the nature of the dependencies.

7. Unary Minus (Negative Numbers)

7.1 Test case identifier

ID: TC-TOK-06

7.2 Test item

Tokenizer module. This test checks whether the tokenizer correctly handles a minus sign used as part of a negative number.

7.3 Input specifications

Input: -5 + 3

7.4 Output specifications

Output: <{Number "-5"}, {Plus "+"}, {Number "3"}>

7.5 Environmental needs

Specify any other requirement.

7.6 Special procedural requirements

Describe any special constraints on the test procedures that execute this test case. These constraints may involve special set up, operator intervention, output determination procedures, and special wrap up.

7.7 Intercase dependencies

List the identifiers of test cases that must be executed prior to this test case. Summarize the nature of the dependencies.

8. Empty Input

8.1 Test case identifier

ID: TC-TOK-07

8.2 Test item

Tokenizer module. This test checks whether the tokenizer handles empty input without crashing.

8.3 Input specifications

Input: “”

8.4 Output specifications

Output: <>

8.5 Environmental needs

Specify any other requirement.

8.6 Special procedural requirements

Describe any special constraints on the test procedures that execute this test case. These constraints may involve special set up, operator intervention, output determination procedures, and special wrap up.

8.7 Intercase dependencies

List the identifiers of test cases that must be executed prior to this test case. Summarize the nature of the dependencies.

9. Malformed Number

9.1 Test case identifier

ID: TC-TOK-08

9.2 Test item

Tokenizer module. This test checks whether the tokenizer detects invalid number formats, such as a number with multiple decimal points.

9.3 Input specifications

Input: 3.1.4 + 2

9.4 Output specifications

Output: Error: Invalid number "3.1.4"

9.5 Environmental needs

Specify any other requirement.

9.6 Special procedural requirements

Describe any special constraints on the test procedures that execute this test case. These constraints may involve special set up, operator intervention, output determination procedures, and special wrap up.

9.7 Intercase dependencies

List the identifiers of test cases that must be executed prior to this test case. Summarize the nature of the dependencies.

10. Consecutive Operators

10.1 Test case identifier

ID: TC-TOK-09

10.2 Test item

Tokenizer module. This test checks whether the tokenizer or validation stage detects invalid consecutive operators.

10.3 Input specifications

Input: 5 ++ 3

10.4 Output specifications

Output: Syntax error at position 3: unexpected operator placement

10.5 Environmental needs

Specify any other requirement.

10.6 Special procedural requirements

Describe any special constraints on the test procedures that execute this test case. These constraints may involve special set up, operator intervention, output determination procedures, and special wrap up.

10.7 Intercase dependencies

List the identifiers of test cases that must be executed prior to this test case. Summarize the nature of the dependencies.

11. Parentheses Tokenization (No Validation)

11.1 Test case identifier

ID: TC-TOK-10

11.2 Test item

Tokenizer module. This test checks whether the tokenizer correctly creates tokens for parentheses without checking whether the parentheses are balanced.

11.3 Input specifications

Input: (2 + 3

11.4 Output specifications

Output: <{LeftParen "("}, {Number "2"}, {Plus "+"}, {Number "3"}>

11.5 Environmental needs

Specify any other requirement.

11.6 Special procedural requirements

Describe any special constraints on the test procedures that execute this test case. These constraints may involve special set up, operator intervention, output determination procedures, and special wrap up.

11.7 Intercase dependencies

List the identifiers of test cases that must be executed prior to this test case. Summarize the nature of the dependencies.